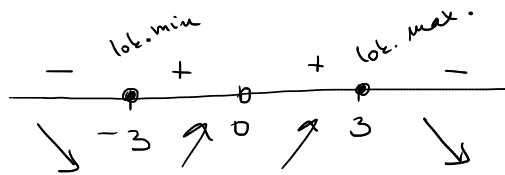


Bonus 4 - REŠENÍ

① $y' = \frac{(3-x)(3+x)}{x^4}$

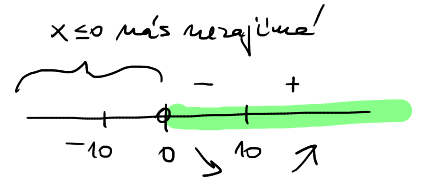


② $S = x \cdot y = 100 \Rightarrow y = \frac{100}{x}$

$l = 2x + 2y = 2x + \frac{200}{x} \rightarrow \text{min}$

$\frac{dl}{dx} = 2 - \frac{200}{x^2} = 2 \left(1 - \frac{100}{x^2}\right) = 2 \left(1 - \frac{10}{x}\right) \left(1 + \frac{10}{x}\right)$

$\underline{x = 10} \Rightarrow \underline{y = 10}$



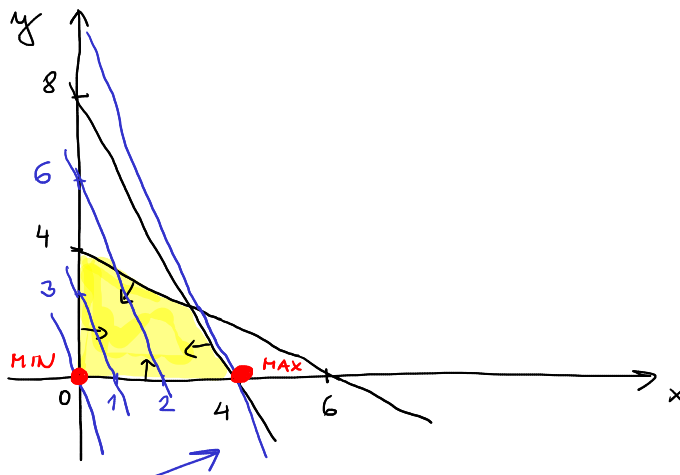
② $z = 3x + y$, $\text{maximize: } 2x + y \leq 8$
 $2x + 3y \leq 12$
 $x \geq 0, y \geq 0$

VRSTEVNICE:

$3x + y = c$

$c = 3: 3x + y = 3$

$c = 6: 3x + y = 6$



maximum v bodě $[4, 0]$, $z_{\max} = 12$

minimum v bodě $[0, 0]$, $z_{\min} = 0$